

Residues

In agriculture, the term ‘residue’ is generally used to describe the small amounts of agricultural and veterinary chemicals, or their breakdown products, that remain in or on an agricultural product.

In the broader context of food safety, a substance can be defined as a ‘contaminant’ if it is ‘an impurity which gives consumers health, safety or cleanliness concerns’.

Contaminants include:

- residues arising from the use of pesticides and veterinary medicines
- heavy metals (e.g. mercury, cadmium, lead)
- naturally occurring chemicals such as mycotoxins (toxins produced by certain fungi)
- microorganisms.

All of these may be present in food, either through natural circumstances or as a consequence of industrial or agricultural activities.

Chemicals that may be detected as residues

These include:

- antibiotics used to control bacterial diseases in animals
- anthelmintics used to control internal parasites in animals
- fungicides used to control fungal diseases in plants and plant products
- insecticides used to control insect pests in crops, protect stored grain and control external parasites on animals
- herbicides used to control weeds in crops
- fumigants used to protect grain and sterilise soil, sheds and bee hives
- hormonal growth promotants used as veterinary medicines or to improve growth in livestock.

The results of NRS residue monitoring program reflect the patterns of use for registered pesticides and veterinary medicines throughout Australia.

Residue testing

The NRS designs residue monitoring program in consultation with industry and the department’s Exports Division. This includes consulting on:

- sampling rates
- selecting chemical—commodity combinations
- designing and managing sampling procedures, including sample collection, identification and dispatch to laboratories

- procuring appropriate analytical services from contract laboratories and monitoring their proficiency
- managing and analysing data
- initiating and collating traceback activities
- managing financial information.

The NRS contracts selected laboratories to undertake necessary chemical analysis. Laboratories are required to demonstrate proficiency in their ability to detect and quantify nominated residues. Selection via competitive tender process is determined on the basis of proficiency and value for money.

The proficiency of participating laboratories is regularly assessed by the NRS to satisfy ongoing competency requirements, and to ensure a high level of confidence in analytical results.

Choosing chemicals and commodities for residue testing

Most agricultural and animal production systems depend on a range of pesticides and veterinary medicines and some production systems may become exposed to environmental contaminants. Current analytical technology can detect chemicals at extremely low concentrations—normally at levels consistent with the chemical use pattern which can indicate whether or not good agricultural practice (GAP) was used or, in the case of exposure to environmental contaminants, whether good control or mitigation arrangements are in place.

Determining which chemicals are monitored in particular commodities is done on the basis of risk profiles which consider:

- Australian Standards for residues, and market access requirements of trading partners
- likelihood of residues occurring in the product (potential for misuse; persistence in the crop, animal or environment; extent of use; use patterns)
- extent and results of previous monitoring for the chemical–commodity combination
- availability of suitable sampling and analytical methods
- international and domestic perceptions of the chemical–commodity combination as a possible public health hazard.

The combinations considered to be the highest risk are included in NRS residue monitoring programmes.

Importing countries sometimes require analyses for particular chemicals of concern in their country.

Consequently, NRS residue monitoring programmes may test for chemicals that are not registered for use in Australia.

Maximum residue limits

The [Australian Pesticides and Veterinary Medicines Authority \(APVMA\)](#) sets MRLs for agricultural and veterinary chemicals registered for use in Australia. These MRLs are set at levels that pose no risk to human health and are not likely to be exceeded if used in accordance with directions on the label.

The APVMA prepares the MRL Standard, which is a document that lists the MRLs for all animal and plant commodities. Following this, the [Food Standards Australia New Zealand \(FSANZ\)](#) and the [Food Ministers Meeting](#) to consider and endorse the MRLs, which are then adopted into the ANZFSO.

Australian Standards

In Australia, FSANZ and the ANZFSO set the residue standards, and the standards are published in the [Australia New Zealand Food Standards Code \(ANZFSO\)](#).

There are three standards used in Australia for residues:

- maximum residue limits (MRLs)
- extraneous residue limits (ERLs)
- maximum levels (MLs).

Standards for Australia's trading partners may vary from country to country, often as a result of different climates, pests and diseases.

The NRS checks the results of sample analysis for compliance with Australian and any relevant international standards. A residue or contaminant is classified as 'present' in a sample if its concentration is greater than the limit of reporting (LOR) established for NRS purposes.

Extraneous residue limits

ERLs concern residues originating from environmental sources of pesticides that are no longer registered.

Maximum levels

MLs were established as an effective risk management function for foods which provide a significant contribution to the dietary exposure of a particular contaminant. MLs are set at levels that are consistent with the protection of public health and safety, and are reasonably achievable through sound production and natural resource management practices.

Interpreting the presence of residues in the absence of an Australian Standard

Where no ML has been set for a particular environmental contaminant in a food, producers are expected to keep the level of contaminants as low as reasonably achievable. However, it is accepted that a low level of contamination may be unavoidable.

In the NRS results tables, these terms are used to indicate residue standards that have not been established for a specific chemical–material combination:

- ‘Not set’ is used for residues of pesticides and veterinary medicines in an edible matrix. It indicates that no standard has been set for the chemical in the particular edible matrix, and any detection is a contravention of the ANZFSC
- ‘Not defined’ is used for residues of pesticides and veterinary medicines in a non-edible matrix. It indicates that due to the inedible nature of the matrix, no standard has been set. Inedible matrices such as urine and faeces may be tested as indicators
- ‘No limit’ is used for an environmental contaminant in an edible matrix. It indicates that no standard has been set for the contaminant. The ‘as low as reasonably achievable’ principle applies, and detections at low levels are allowable.

Testing programs

NRS residue testing programs involve random and targeted monitoring of animal and plant products. Laboratory performance evaluation and proficiency testing ensure the reliability of the analytical results upon which the residue testing program depend.

Random monitoring

Programmes are designed to estimate the occurrence of a residue(s) by using randomised sampling processes. NRS random residue monitoring data facilitates and underpins the demonstration of:

- compliance with requirements for domestic consumption
- the certification of commodities for export (where required)
- the setting or review of Australian Food Standards.

This underpinning helps participating industries to maintain access to important export markets and a competitive advantage in those markets. These programmes also play an important role in the negotiation of access to new and potential markets.

Targeted monitoring

Targeted monitoring program run in conjunction with random sampling arrangements and are designed to obtain more focused information concerning known or potential residue problems. These programmes use specific sampling processes tailored to the particular area or participants of concern.

Traceback

Laboratories test samples against an agreed chemical screen which meets market requirement. If a laboratory finds a sample that contains a residue above the Australian Standard, a traceback investigation is undertaken to establish the cause. The responsible state or territory agency then

provides advice to the producer to prevent recurrence. In more serious circumstances, regulatory action may also be taken.

All traceback activities and findings are reported to the NRS. This feedback is important in highlighting potential problems, such as inappropriate chemical use and improving farm practices. Where appropriate, traceback information is also forwarded to industry and government authorities for consideration. Traceback information may also be forwarded to the APVMA for consideration during its chemical review processes.

Source: <https://www.agriculture.gov.au/agriculture-land/farm-food-drought/food/nrs/about-residues-and-residue-testing>